

Experiment 3 — FPGA-Based Smart Industrial Machine Health Monitor

RTL module: Industrial_Machine_Health Testbench: tb_Industrial_Machine_Health Simulation duration: 70 ns

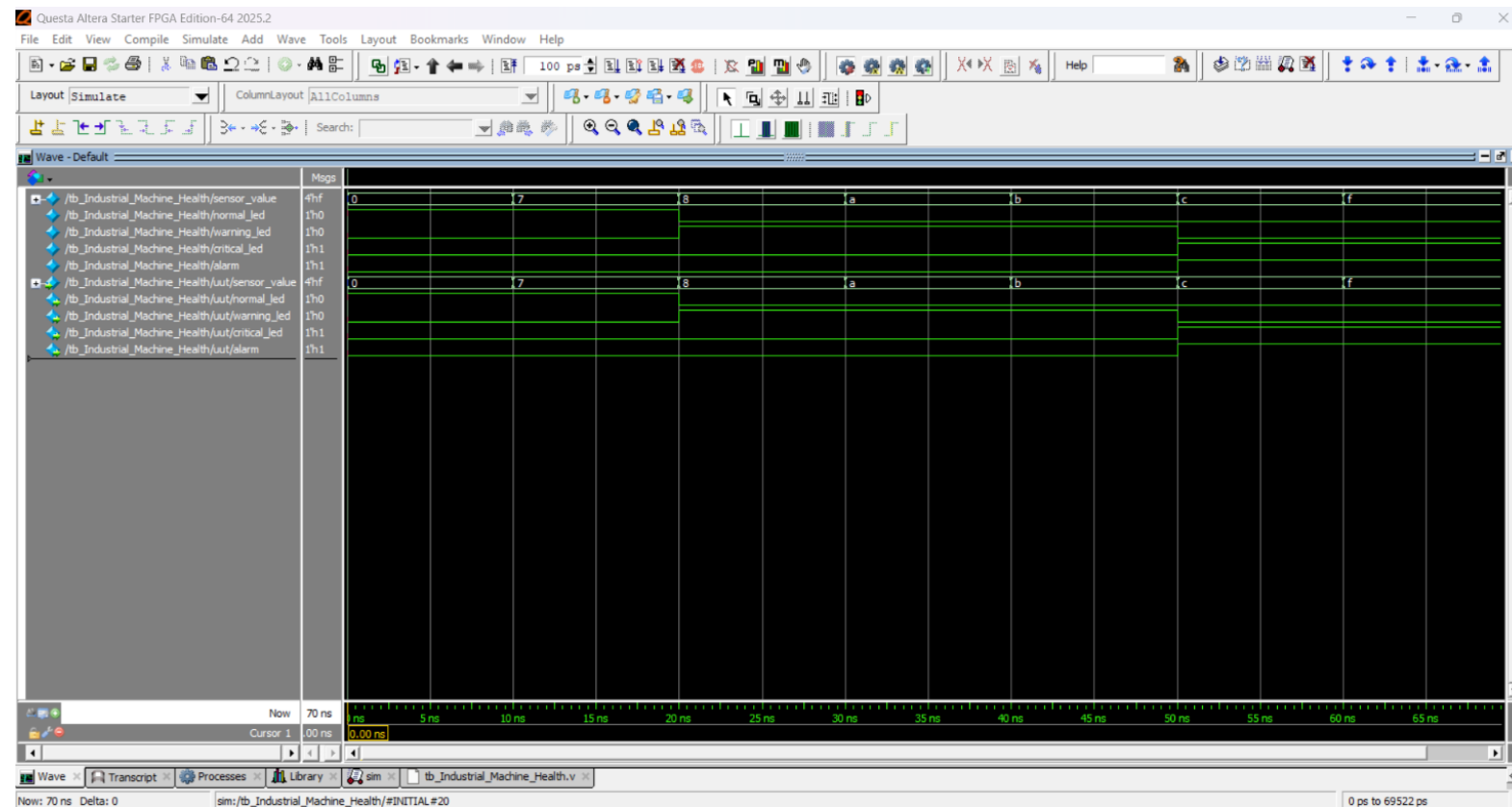
Simulation Objective

The simulation verifies three machine-health ranges: normal for sensor values 0–7, warning for 8–11, and critical/alarm for 12–15.

Verification Sequence

Time	Input	Condition	Out-1	Out-2	Out-3	Out-4
0–10 ns	0	Normal	1	0	0	0
10–20 ns	7	Normal	1	0	0	0
20–30 ns	8	Warning	0	1	0	0
30–40 ns	10	Warning	0	1	0	0
40–50 ns	11	Warning	0	1	0	0
50–60 ns	12	Critical	0	0	1	1
60–70 ns	15	Critical	0	0	1	1

Waveform Evidence — The screenshot below is the captured Questa waveform for this experiment.



Result: The waveform completes the 70 ns test sequence and demonstrates the expected logical transitions for sensor_value, normal_led, warning_led, critical_led, alarm. The simulation stops normally at the end of the testbench.

Conclusion

Simulation verification completed successfully with the DUT and corresponding testbench loaded in Questa. The observed transitions are consistent with the intended RTL logic.